

Collocation Engine

Ask one point of the ocean what every instrument says about it - and how far away each of those instruments actually was.

PORT	PAGE FILE	MODULE	TESTS
8502	app/phase2/collocation_page.py	phase2/data/collocation.py	test_collocation.py

Every other feature in this project eventually asks the same question: what did each source report here? Answering it once, in one place, with one set of rules, is what stops five different features from quietly inventing five slightly different answers.

1. What this feature is, in plain words

You give it a latitude, a longitude and a date. It returns a single record listing what the satellite saw, what the GLORYS reanalysis says, what our model reconstructed, what the wind was doing, and - if a real Argo float happened to be nearby - what that float actually measured in the water.

The part that makes it trustworthy is the second half of every record: **how far away each match really was**, in kilometres and in days. A float 2 km away and on the same day is strong evidence. A float 60 km away and four days later is weak evidence. Both are useful; pretending they are the same is not.

So the engine never hands you a bare label. It measures the offsets first and derives the quality from them: HIGH within 2 days, MEDIUM within 5, LOW within 10, and REJECT beyond that or outside the grid. You can always see the number the label came from.

THE EVERYDAY PICTURE

It is the ocean's version of a witness statement that also records where the witness was standing. Two people can both say they saw the accident - one from the pavement, one from four streets away. A serious investigator writes down the distance.

2. What you actually see on the screen

One point in, one complete evidence card out:

- The value each source reports at that point, side by side, each keeping its own numbers verbatim - no source is ever overwritten or substituted for another.
- The measured spatial offset in kilometres and the temporal offset in days, for every match.
- A colour-coded quality flag - green HIGH, amber MEDIUM or LOW, red REJECT - derived from those offsets, never asserted.
- An era switch: the Phase-1 record of 48 monthly fields (2019-2022), or the Phase-2 record of 388 consecutive daily fields (1 June 2025 to 23 June 2026).
- Full provenance: which files were read, which grid cell was used, which offsets applied.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/collocation_page.py --server.port 8502
```

- 1 Run the command above and open **http://localhost:8502**.
- 2 Choose the era. Pick **daily** to work in the shipped model's own time window, or **phase1** to reproduce the originally published Phase-1 collocation numbers.
- 3 Type a latitude and longitude inside the basin - 15.0 N, 65.0 E in the Arabian Sea is a good demonstration point.
- 4 Pick a date inside the chosen era's range.
- 5 Read the quality flag first, then the offsets, then the values. That order is deliberate: it teaches the audience to check the evidence before reading the number.
- 6 Try a point on land, such as 21.75 N, 47.50 E, to see the engine refuse rather than invent a value.

AN EMPTY RESULT IS A REAL ANSWER

If no float was near that point on that date, the engine reports a gap in the float table - not an empty ocean. Those are completely different claims, and confusing them once cost this project a wrong conclusion, which is recorded in the corrections log.

4. Why the problem statement needs this

The problem statement requires preprocessing and harmonisation of multiple data sources, and independent validation against float observations. Both of those are collocation problems: you cannot harmonise or validate anything until you can say, defensibly, that two measurements refer to the same piece of ocean at the same time.

It also protects every downstream number. The validation lab, the Argo overlay, the transect and the cyclone study all consume this engine rather than writing their own matching rule, so a single, reviewable definition of 'nearby' governs the whole project.

PS requirement	How this feature answers it
Preprocessing and harmonisation	Seven channels from five distinct products answered at one point, each keeping its own units and provenance.
Independent validation with Argo	Provides the matched float pairs every validation surface in the project uses.
Report bias honestly	Records the offsets that any bias estimate depends on, so a reviewer can weigh the match quality themselves.

5. The measured numbers behind it

The design was forced by measurements taken on 26 August 2026, not chosen by preference [VERIFIED]:

Measurement	Result	What it forced
Distance from an Argo float to the nearest grid centre	median 10.8 km, 95th percentile 16.1 km, max 19.0 km	Space is easy - a 0.25 degree cell is about 27 km wide, so nearest-neighbour is always inside half a cell.
Time from a float to the nearest gridded date (2,455 profiles)	within 1 day: 9.5% / 3 days: 22.6% / 5 days: 36.5% / 7 days: 50.3% / 15 days: 99.7%	Time is the hard decision. Tighter matching means less data; looser matching means weaker evidence. There is no free choice.
Chosen thresholds	HIGH 2 days, MEDIUM 5 days, LOW 10 days, reject beyond 10 days or beyond 20 km	Stated openly and configurable, so a reviewer can move them and re-run.

- Because the offsets are recorded on every record rather than averaged away, any later result can be re-checked at a stricter tolerance. That is how the project separated real reanalysis error from collocation mismatch in Feature 3.

6. The impact - and who it is for

Who	What changes for them
Ocean scientists	A single defensible answer to 'do these two measurements describe the same water', instead of a per-study convention.
The jury	Direct evidence that our validation is not cherry-picked - the match quality is on screen beside every comparison.
INCOIS / MoES	A reusable harmonisation layer. Any new instrument - a buoy, a glider, a new satellite - plugs in as another source with its own measured offsets.
The team	One matching rule to review instead of five, and therefore one place a mistake could hide instead of five.

WHY SOCIETY GAINS FROM THIS

Ocean data arrives from instruments that never agree on where or when to measure. A public agency merging them for a public forecast needs the merge itself to be auditable. This is that audit trail, and it refuses to hide a weak match behind a confident label.

7. Say this to the jury

Ask it about any point in the basin and it tells you what every instrument says there - and, crucially, how far each of those instruments actually was in kilometres and days. We measured that a random float sits a median 10.8 km from a grid centre but a median 7 days from a gridded date, so space is easy and time is the real scientific decision. Rather than pick a tolerance silently, we record the offset on every single record and derive the quality from it.

8. Questions the jury will ask

Q. Why not just use a weighted score combining distance and time?

A. Because there is no defensible exchange rate between kilometres and days, and inventing one would bury an assumption inside a number. We rank by distance first and use time to break ties, and we show both.

Q. What happens if no float is nearby?

A. The record says the float table has no rows for that point and date. It never returns zero, and it never returns a neighbour's value. In this project an absence is reported as an absence.

Q. Could a bad match still slip into your headline result?

A. The headline validation uses a 5-day tolerance and the record carries the collocation statistics - max 18.7 km, mean 10.5 km. The reanalysis comparison was also re-run at a tighter 3-day, 25-km match and the numbers barely moved, which is how we know the error we report is real error and not mismatch.

9. The honest limit - say it before they find it

IT CANNOT FIX SPARSE DATA

Collocation makes match quality visible; it does not create floats where there are none. In the Phase-1 era the float table holds 2022 rows only, so a query in 2019 correctly returns an empty match - and reading that as 'the ocean was empty' rather than 'the table has no rows' is exactly the error this engine now blocks by reporting coverage explicitly.

IN ONE LINE

One point of ocean, every source's answer, and the measured distance in kilometres and days behind each of those answers.

10.8 km

MEDIAN FLOAT-TO-GRID DISTANCE

7 days

MEDIAN FLOAT-TO-DATE OFFSET

4 flags

HIGH / MEDIUM / LOW / REJECT

Feature 2 of 17. Next: Feature 3.